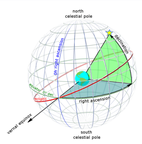
Funded by the U.S. National Science Foundation and the Simons Foundation

When Geometry Meets Causality:
Statistical Methods for Cosmic Web Analysis and Beyond

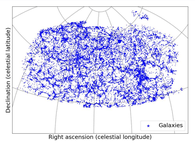
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Three Statistical Challenges Behind Cosmic Web Analysis

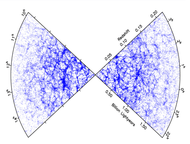
1. Nonlinear Geometry and Low-Dimensional Structures: $\Omega_q = \{x \in \mathbb{R}^{q+1} : ||x||_2 = 1\}$.



Celestial sphere Ω_2.

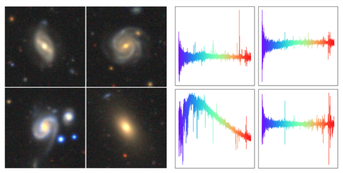


Galaxies on Ω_2 (fixed redshift).

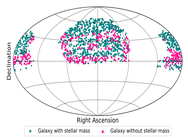


Galaxies across redshifts $\Omega_2 \times \mathbb{R}$.

2. High Dimensionality and Missingness: $X \in \mathbb{R}^d$, $d \gg n$, and $Y \in \mathbb{R} \cup \{\texttt{NA}\}$.

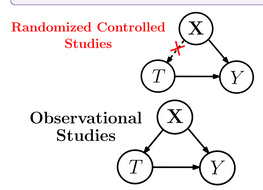


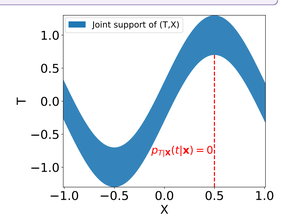
Photometric and spectral features.



Galaxies in a high redshift slice.

3. Confounding and Positivity Violations: $p_{T|X}(t \mid x) = 0$ for some $(t, x) \in \mathcal{T} \times \mathcal{X}$.

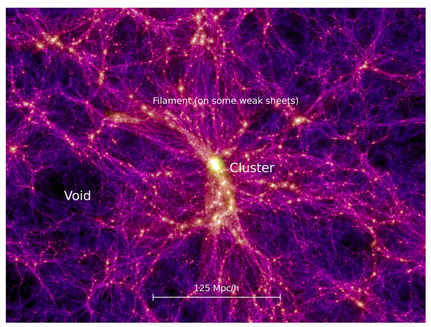




Main Theme: Develop **geometry-aware** statistical learning and causal inference methods to address these challenges.

Geometry-Aware Cosmic Web Detection

Goal: Detect cosmic filaments and nodes while respecting the curvature of the sky.



Cosmic Web: The large-scale network structure revealing the non-uniformity of matter distribution in the Universe.

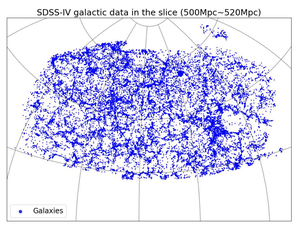
- 0D **galaxy clusters** (*cosmic nodes*);
- 1D **filaments** connecting clusters;
- 2D tenuous **sheets** or **walls**;
- 3D nearly empty **cosmic voids**.

Cosmic Filament Model: Represent the 1D filament as directional density ridges of the galaxy density function f on Ω_2:

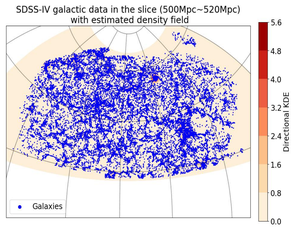
$$R_1(f) = \left\{ x \in \Omega_2 : v_2(x)^T \operatorname{grad} f(x) = 0, \lambda_2(x) < 0 \right\},$$

where $\operatorname{grad} f(x)$ is the *Riemannian gradient* of f at $x \in \Omega_2$, and $v_2(x)$ is the tangent-space eigenvector of the *Riemannian Hessian* $\mathcal{H}f(x)$ corresponding to its smaller eigenvalue $\lambda_2(x)$.

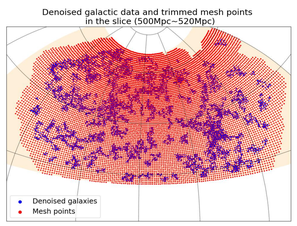
Cosmic Web Detection in an Example Redshift Slice



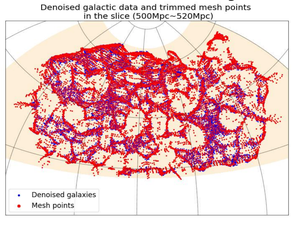
Step 1: Slice the Universe and select the North Galactic Cap.



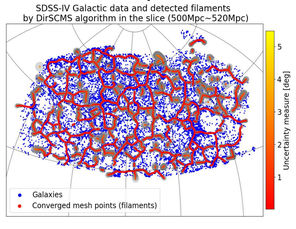
Step 2: Estimate the galaxy density field.



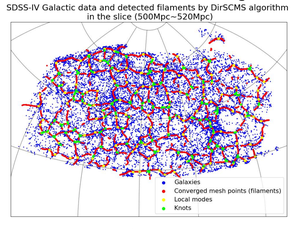
Step 3: Denoise the density estimate and initialize mesh points.



Step 4: Apply our DirSCMS algorithm to mesh points.



Step 5: Run DirSCMS until converged and quantify uncertainty.



Step 6: Identify local modes and filament knots.

Output: A public cosmic web catalog across **325 redshift slices** of thickness 20 Mpc.

- *Full catalog:* `doi.org/10.5281/zenodo.6244866`.
- *Python package:* `pypi.org/project/sconce-scms/`

SCONCE SCMS

Inference on the Cosmic Web

Question: Are galaxies farther from filaments *significantly less massive* after adjustment for observed features?

Challenges and Assumptions

1. Missing Outcomes: Stellar mass is unobserved for some galaxies.

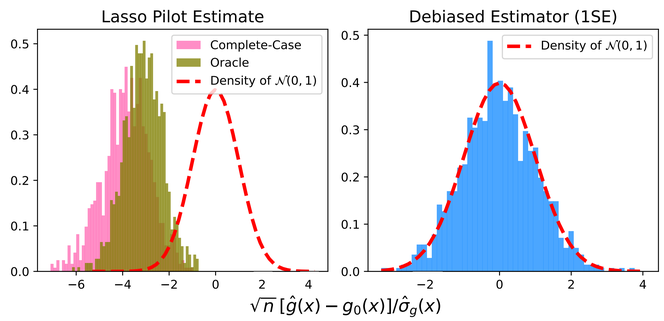
- *Missing at random:* $Y \perp\!\!\!\perp R \mid (D_{\text{fila}}, X)$, where $R = 1$ if Y is observed and $R = 0$ otherwise.

2. High-dimensional Covariates: Each galaxy has many observed features and nonlinear transformations.

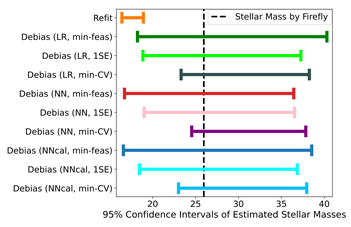
- *Sparse linear model:* $Y = \beta_0 D_{\text{fila}} + \beta_1^T X + \epsilon$, with $||\beta_1||_0 = s_\beta \ll d$, $\mathbb{E}(\epsilon \mid D_{\text{fila}}, X) = 0$, and $\operatorname{Var}(\epsilon \mid D_{\text{fila}}, X) = \sigma_\epsilon^2$.

Debiased Inference Procedure

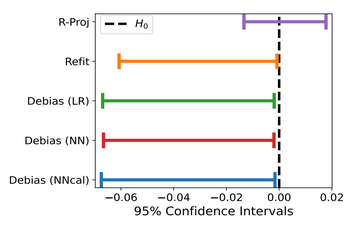
1. Compute a *Lasso pilot estimate* with complete-case data.
2. Estimate the *observation probabilities* given the covariates.
3. Solve a *convex debiasing program* for balancing weights.
4. Derive the *debiased estimator* of β_0 with a weighted residual correction and construct its confidence interval.



Inference for Galaxy Properties



95% CIs for conditional mean stellar mass.



95% CIs for the distance coefficient β_0 .

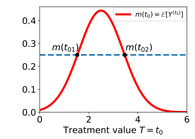
Finding: The proposed 95% confidence intervals (CIs) for β_0 lie below zero: galaxies farther from nearby filaments tend to have lower stellar masses after adjustment.

Causal Inference Under Positivity Violations

- *Treatment* $T \in \mathcal{T} \subset \mathbb{R}$*:* Local environmental density of a galaxy.
- *Covariates* $X \in \mathcal{X} \subset \mathbb{R}^d$*:* Observed treatment–outcome confounders.
- *Outcome* $Y \in \mathcal{Y} \subset \mathbb{R}$*:* Stellar mass or another galaxy property.
- *Potential Outcome* $Y^{(t_0)} \in \mathcal{Y}$*:* Outcome if treatment were set to t_0.

Causal Estimands:

- Dose-response curve $t_0 \mapsto m(t_0) := \mathbb{E}[Y^{(t_0)}]$;
- Derivative effect curve $t_0 \mapsto \theta(t_0) = m'(t_0) = \frac{d}{dt}\mathbb{E}[Y^{(t_0)}]$.

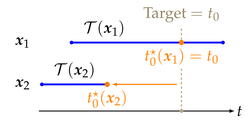


Issue: When positivity fails at t_0 with $p_{T|X}(t_0|x) = 0$, $m(t_0)$ and $\theta(t_0)$ need not be *identifiable* without additional assumptions.

Nearest Feasible Treatment Policy (NFTP)

For the nonempty and closed conditional support $\mathcal{T}(x)$ of $p_{T|X}(t|x)$, assign the nearest feasible level:

$$t_0^\star(x) := \arg\min_{t \in \mathcal{T}(x)} |t - t_0|.$$



***NFTP causal estimand*:** let $\mu(t, x) = \mathbb{E}(Y|T = t, X = x)$, and

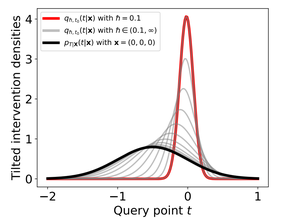
$$\bar{m}_0(t_0) := \mathbb{E}\left[Y^{(t_0^\star)}\right] \overset{(\blacktriangle)}{=} \int_{\mathcal{X}(t_0)} \mu(t_0, x)\, d\mathrm{P}_X(x) + \int_{\mathcal{X}\setminus\mathcal{X}(t_0)} \mu(t_0^\star(x), x)\, d\mathrm{P}_X(x)$$

- $Y^{(t_0^\star)}$: potential outcome under the dynamic treatment regime $x \mapsto t_0^\star(x)$.
- $(\blacktriangle)$: the identification formula without global positivity.
- $\mathcal{X}(t_0) = \{x \in \mathcal{X} : t_0 \in \mathcal{T}(x)\}$, and $\bar{m}_0(t_0) = m(t_0)$ under positivity.

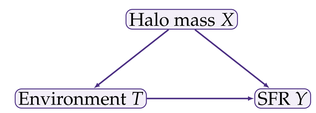
Approach the NFTP intervention through a positivity-lean point-targeted (PLPT) tilted intervention:

$$q_{\hbar,t_0}(t|x) := \frac{\eta(t; t_0, \hbar)\, p_{T|X}(t|x)}{\int_{\mathcal{T}} \eta(u; t_0, \hbar)\, p_{T|X}(u|x)\, du}.$$

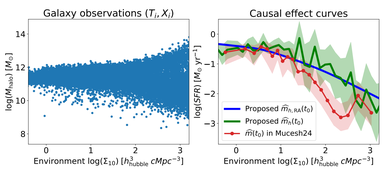
- $t \mapsto \eta(t; t_0, \hbar) \propto \exp\left[-\left|\frac{t-t_0}{\hbar}\right|^\gamma\right]$ for some $\gamma > 0$;
- $Q_{\hbar,t_0}(\cdot \mid x) \Rightarrow \delta_{t_0^\star(x)}(\cdot)$ as $\hbar \to 0$.



"Nature vs. Nurture" in Galaxy Evolution



- **Data:** 18,629 galaxies with $M_\star > 10^9 M_\odot$ from IllustrisTNG (TNG100), with redshift $z \approx 0$–3.
- T: 10-th nearest neighbor density.
- Y: Instantaneous SFR for each galaxy.
- X: Halo mass, $\log(M_{\text{halo}})$.



Finding: After adjusting for halo mass (*nature*), the environment effect (*nurture*)

- is strongly negative at low z,
- weakens for $z > 0.5$, and
- shows no clear positive trend at high z.

